

Recasting the Model in Terms of Visits

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Recasting the model to focus on visits to shops

Household i chooses visits v_i to max. utility
st. budget constraint, taking as given
tightness α & price p .

Household's problem

- consumption: $c_i = y_i - p \cdot v_i = [q(x) - p] \cdot v_i$
- spending/output: $y_i = q(x) \cdot v_i$

$$\max_{v_i} \frac{\alpha}{1+\alpha} [(q(x) - p) v_i]^{\frac{\alpha-1}{\alpha}} + \frac{1}{1+\alpha} \left(\frac{m_i}{p} \right)^{\frac{\alpha-1}{\alpha}}$$

$$\text{st. } m_i + p \cdot q(x) v_i = \mu_i + p f(x) k_i$$

$$\hookrightarrow \max_{v_i} \frac{\alpha}{1+\alpha} [(q(x) - p) v_i]^{\frac{\alpha-1}{\alpha}} + \frac{1}{1+\alpha} \left[-q(x) v_i + \frac{\mu_i}{p} + f(x) k_i \right]^{\frac{\alpha-1}{\alpha}}$$

concave maximization problem

$$\text{FOC: } \frac{\alpha}{1+\alpha} \cdot \frac{\alpha-1}{\alpha} [q(x) - p]^{\frac{\alpha-1}{\alpha}} v_i^{\frac{\alpha-1}{\alpha}} = \frac{q(x)}{1+\alpha} \cdot \frac{\alpha-1}{\alpha} \left[\dots \right]^{\frac{\alpha-1}{\alpha}} = 0$$

$$v_i = \frac{\alpha}{q(x)^\alpha} [q(x) - p]^{\alpha-1} [f(x) k_i + \frac{\mu_i}{p} - q(x) v_i]$$

$$\left[1 + \frac{\alpha}{q(x)^\alpha} [q(x) - p]^{\alpha-1} \right] \cdot v_i = \frac{\alpha}{q(x)^\alpha} [q(x) - p]^{\alpha-1} \left[f(x) k_i + \frac{\mu_i}{p} \right]$$

$$V_i(x, p) = \frac{x^\xi (q(x) - p)^{\xi-1} q(x)^{1-\xi}}{1 + x^\xi (q(x) - p)^{\xi-1} q(x)^{1-\xi}} \left[f(x) k_i + \frac{w_i}{p} \right]$$

$$q(x) \sigma_i(x, p) = \frac{x^\xi (q(x) - p)^{\xi-1} q(x)^{1-\xi}}{1 + x^\xi (q(x) - p)^{\xi-1} q(x)^{1-\xi}} \left[y_{x, k_i} + \frac{w_i}{p} \right]$$

$\uparrow$
 $y_i(x, p)$

$\uparrow$
 $\sigma_i(x) = \text{MPS}$

$\uparrow$
[initial wealth + income]

Aggregate number of orbits $V(x, p) = \sum_i V_i(x, p)$

$$q(x) \sigma(x, p) = \sigma(x) \left[f(x) \cdot k + \frac{w}{p} \right]$$

$\downarrow$
same as AD curve